2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	4
Project Name	AI Music Mood Classifier
Report Date	Sep 16, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

This week, I have properly set up my data_loader.py and features.py classes, along with using them in my main.py file. The data_loader class loads the annotation and audio files that are to be processed by the program, and the features class extracts the MFCC, chroma, spectral centroid, spectral bandwidth, spectral contrast, spectral rolloff, zero-crossing rate, and tempo features of each song put into the AI Music Mood Classifier. The main file utilizes them in order to load the annotations and audio files that are to be classified and extracts the features from them. It also saves the extracted features so larger datasets can be cached and do not need to have each annotation and audio file run through the classifier again. The software specification requirements and the software design document have also had their first draft completed.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

The only main blockers I have been having are the size of the dataset, causing my program to take a while to run, which means that errors from typos and other various issues take longer to find.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

My goals for next week are to test around with implementing my baseline classifier, as I need to start training my classifier whenever I can, so it eases the burden of the project later down the road.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

For the upcoming baseline classifier, is it better to start with something very lightweight (like logistic regression) or jump to more flexible models (random forest, SVM) right away? Should I separate feature caching into its own script (precompute_features.py) or integrate it directly into main.py so the workflow stays simple?

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
9/9 8:00	9/9 12:00	4	Consulting about what features I should extract from the audio files and about what the librosa feature extraction function does and what inputs it takes. Built skeleton of main.py, data_loader.py, features.py, and created prepare_deam_annotations file that cleaned up and put both annotations from the DEAM dataset into one csv annotation file for ease of use. Also added scatterplot visualization of the data via matplotlib.
9/10 16:00	9/10 19:00	3	Building up from the skeleton and making sure that it extracts the mfcc from the audio files.
9/12 16:00	9/12 21:00	5	Watched many youtube tutorials on librosa, pytorch, and audio classification with pytorch.
9/14 10:00	9/14 13:00	3	Researched pytorch some more through youtube tutorials and documentation from different audio classification projects.
9/15 14:45	9/15 17:00	2.25	Added chroma, centroid, bandwidth, contrast, rolloff, zero-crossing rate, and tempo features to be extracted via features.py. I also cleaned up my main.py file and allowed the feature extraction from the dataset to be saved so I won't have to re-extract features from all 1802 files of the dataset each time I run the program.
	WEEKLY TOTAL	17.25	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

So far, I have spent 51 hours total on the project. According to the timeline I built with ChatGPT's help, I am technically ahead of pace. The full feature extraction, caching dataset results from feature extraction, and visualization I added with matplotlib were meant for the week of september 15-21, but I have already implemented those steps.

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE